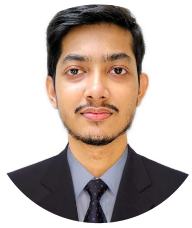
MD MAHIR UDDIN



AI/ML Engineer | Computer Vision & Gen AI

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Portfolio | LinkedIn | GitHub | Codeforces | LeetCode | Kaggle | Coursera

ABOUT ME

I am an AI/ML Engineer with strong foundations in Mathematics, Statistical Modeling, Deep Learning, Computer Vision, and Gen AI. Driven by a long-standing love for Mathematics and problem-solving, I transitioned into the field of Data Science by pursuing an MSc in CSE (Data Science). Currently working on an ADB-funded project that uses AI and IoT to increase right first time shade in bulk dyeing and reduce resource usage in textile dyeing factories. Worked on object detection, LLM fine-tuning, and RAG, with ongoing research in developing a density-based clustering algorithm and modification of the ADAM optimizer for reduced oscillation and faster convergence.

SKILLS

- **Programming Languages:** Python, SQL, C, C++, JavaScript
- **Machine Learning & AI:** Machine Learning, Deep Learning, Computer Vision, Generative AI
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, LangChain
- **MLOps/Deployment:** FastAPI, Docker, AWS EC2, Google Cloud Run
- **Others:** Git, Linux, HTML, CSS, MS Excel, LaTeX
- **Additional Knowledge:** Big Data, Cloud Computing, Data Engineering

PROJECTS

Diffusion-based Virtual Try-On System

Stable Diffusion, Hugging Face, FastAPI, Docker, GCP

- Built an end-to-end virtual try-on backend using diffusion models to generate realistic garment transfers from person and clothing images.
- Designed a Dockerized FastAPI inference service with GPU acceleration, enabling scalable cloud deployment on GCP.

Skin Diseases Prediction with LLM Recommendations

Vision Transformer, LLM, FastAPI, Docker, PyTorch

- Built a Data efficient image Transformer (DeiT)-based skin disease classifier (97%+ accuracy) with real-time inference via FastAPI.
- Integrated LLM(Gemini 2.5 Flash) driven recommendations and deployed using Docker on the cloud with an interactive streamlit interface.

LLM Fine-Tuning with RAG for Investment Guidance

LLM, LangChain, Vector DB, RAG

- Fine-tuned a domain-specific LLM and integrated Retrieval-Augmented Generation to deliver context-aware investment insights.
- Combined model fine-tuning with dynamic retrieval from books, annual reports, and financial articles from the web.

Portrait Image to Sketch Translation with Pix2Pix GAN

TensorFlow, Pix2Pix GAN, U-Net, PatchGAN

- Developed a Pix2Pix conditional GAN with U-Net generator and PatchGAN discriminator for photo-to-sketch translation on FS2K dataset.
- Generated high-fidelity, structurally accurate sketches using adversarial and L1 loss for paired image-to-image translation.

More projects: mahiruddin.vercel.app/projects

ACHIEVEMENTS

- Codeforces max rating **1309**; ranked **723rd** among **20,792** Bangladeshi coders (**Top 2.98%**).
- International Youth Math Challenge (IYMC) 2025 **Gold Medalist**.
- Ranked **14th nationwide** in BUTEX A Unit 2018 admission exam among **13,000+** candidates.
- Ranked **9th nationwide** in Jahangirnagar University H Unit 2018 admission exam among **34,000+** candidates.

RESEARCH

Research works completed as part of coursework and independent study (unpublished; publishable work not included).

Automated Density-Based Splitting of Merged Clusters

Clustering, Unsupervised Learning Paper

- Proposed a clustering refinement framework to detect and split incorrectly merged clusters using density analysis
- Designed a recursive splitting mechanism to dynamically adjust cluster structure without predefined *k*.

Fabric Defect Detection Using Image Processing and CNN

Computer Vision, CNN, ResNet-50 Paper

- Developed a vision-based textile defect detection system combining contrast enhancement with CNNs.
- Improved defect visibility and classification robustness under varying fabric textures and lighting conditions.

Intelligent Irrigation Decision Support System Using IoT and Weather Data

IOT, MQTT, Machine Learning Paper

- Built a data-driven decision support system integrating IoT sensor data with weather information.
- Optimized irrigation planning to reduce water consumption and enhance resource efficiency, with motor operation time and duration controlled via IoT microcontrollers.

EDUCATION

M.Sc. in Computer Science & Engineering

United International University

Nov 2024–July 2026 Dhaka, Bangladesh

- Major: Data Science
- CGPA: 3.82

B.Sc. in Textile Engineering

Bangladesh University of Textiles (BUTEX)

Jan 2018–Oct 2023 Dhaka, Bangladesh

- Specialization: Wet Processing Engineering
- CGPA: 3.34

REFERENCES

Dr. Ohidujjaman

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United International University

- Email: ohidujjaman@cse.uiu.ac.bd
- Phone: +880 179 808 3444

Dr. Mohammad Nurul Huda

Professor, Dept. of CSE
United International University

- Email: mnh@cse.uiu.ac.bd
- Phone: +880 1673-476433

EXPERIENCE

Research Engineer (AI/ML) *(Part-time)*

ADB-Funded AI & IoT Project

Feb 2026 – Present

Dhaka, Bangladesh

Mathematics & Physics Lecturer *(Part-time)*

Sunrise Coaching Centre

Aug 2018 – Sep 2022 (4 yrs 2 months)

Dhaka, Bangladesh

Mathematics Lecturer *(Seasonal)*

UCC (University Coaching Center)

Apr 2019 – Aug 2022 (3 yrs 5 months)

Dhaka, Bangladesh

CERTIFICATIONS

- **Machine Learning Specialization**
  Stanford · Coursera Credentials
- **Deep Learning Specialization**
  DeepLearning.AI · Coursera Credentials
- **IBM Generative AI Engineering Professional Certificate**
  IBM · Coursera Credentials
- **Mathematics for Machine Learning and Data Science**
  DeepLearning.AI · Coursera Credentials